System Design & Wireframing

Report 2

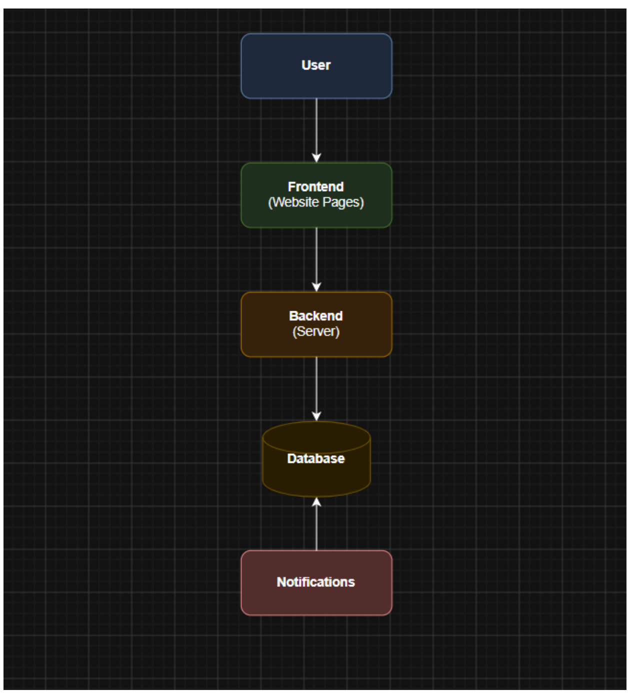
Divija Reddy Varkuti
CSCI 4950_OL1 STEM Project Laboratory
Professor Mauricio Yepez
June 28, 2026

1. System Architecture & Data Design

System Architecture

The Arise application follows a simple web application architecture. Users interact with the website through the frontend, where they can log in, manage tasks, view their calendar, and track wellness activities. The frontend sends requests to the backend, which processes the information, communicates with the database, and returns the requested data. The notification system reminds users about upcoming tasks, events, and medication schedules.

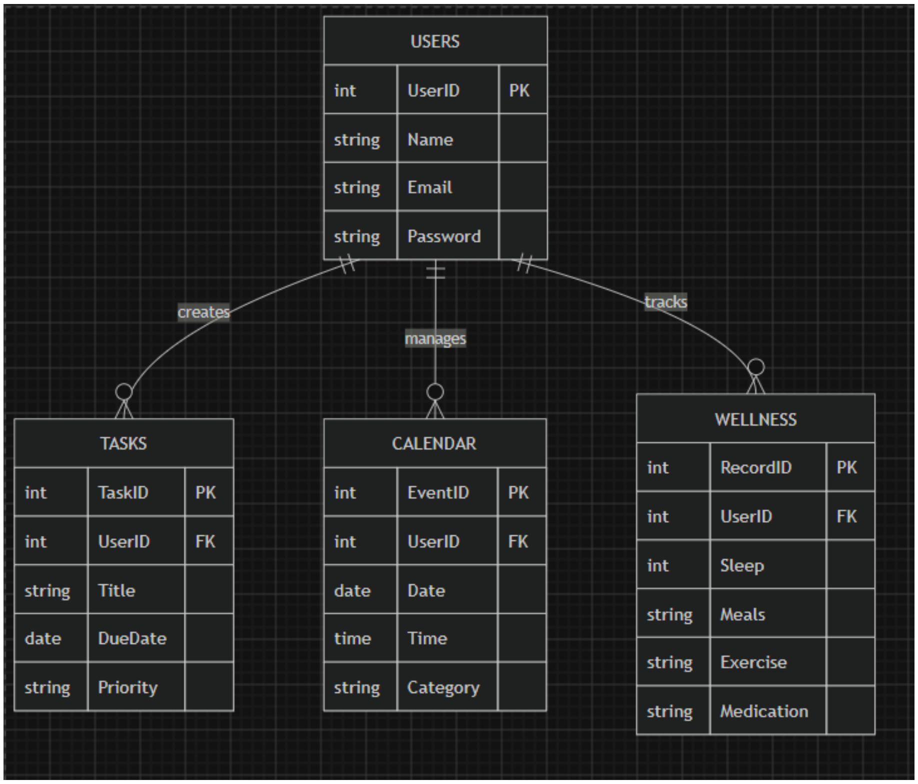
System Architecture Diagram



Data Design

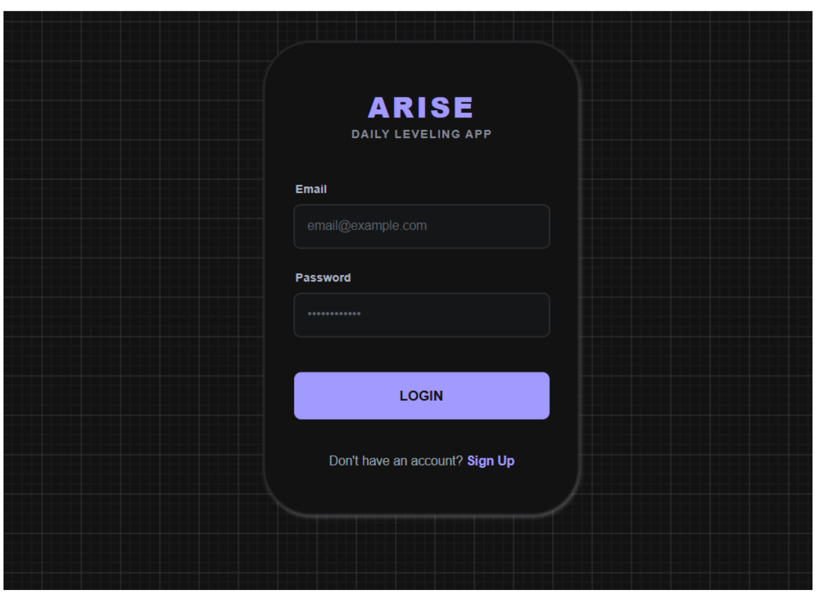
The database stores all information related to each user. Every user has their own tasks, calendar events, and wellness records. The UserID connects each table so that each user's information remains organized and secure.

Database Diagram



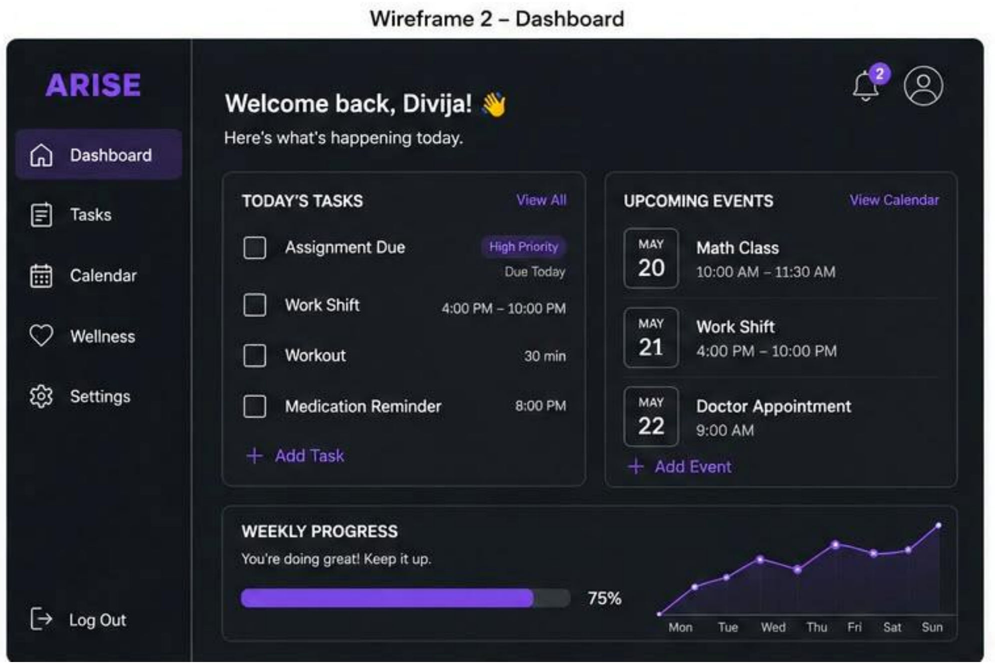
2. Wireframes & Interface Layouts

Wireframe 1 – Login Page



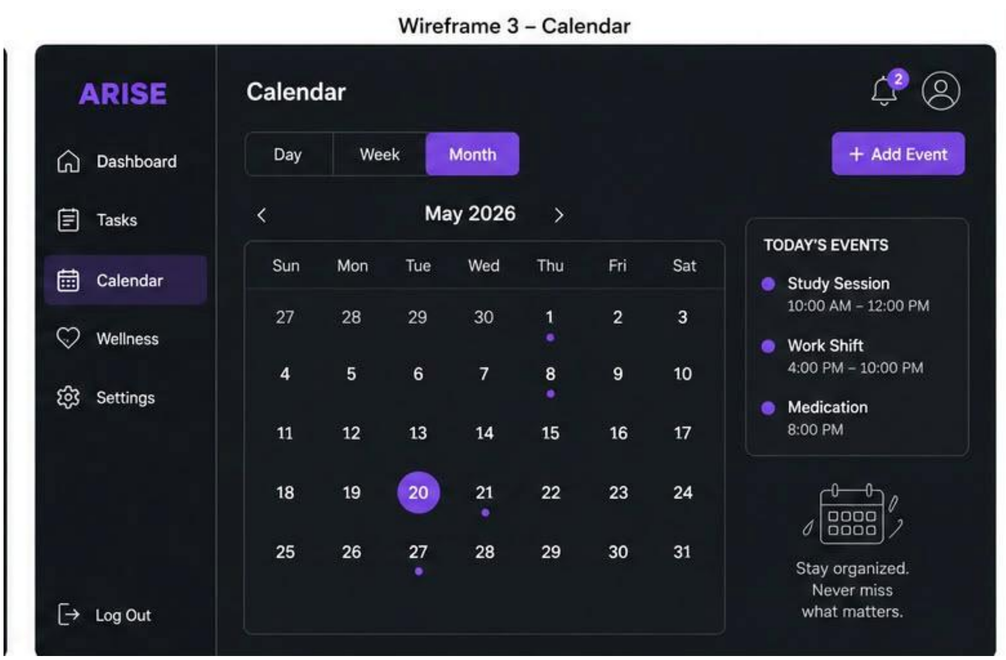
This screen allows users to create an account and securely log into the application. It fulfills Functional Requirements 1 and 2.

Wireframe 2 – Dashboard



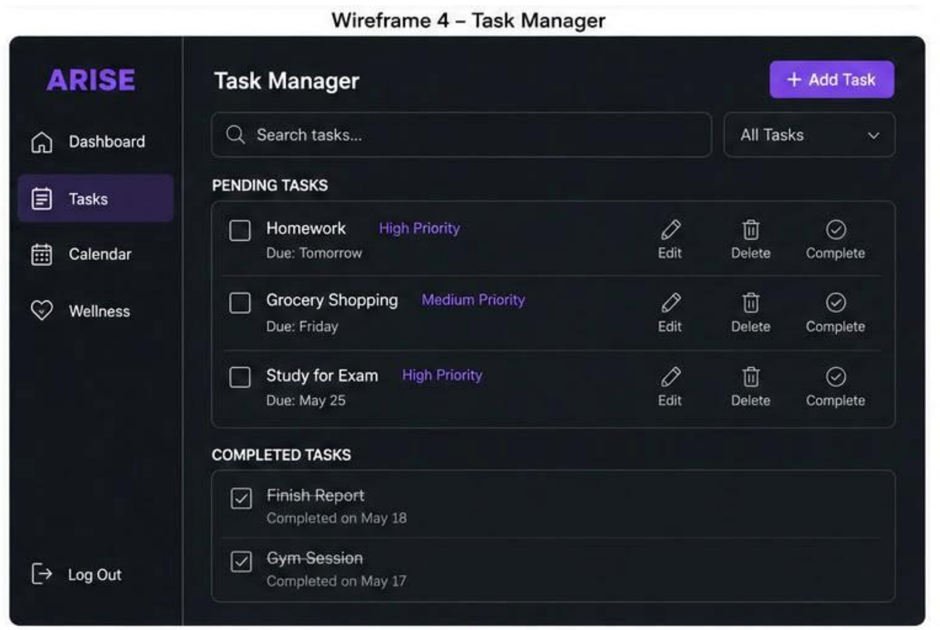
The dashboard provides users with a personalized overview of their daily schedule, reminders, and progress. It fulfills Functional Requirement 3.

Wireframe 3 – Calendar



The calendar allows users to organize classes, work shifts, appointments, and recurring events. It fulfills Functional Requirements 5 and 6.

Wireframe 4 – Task Manager



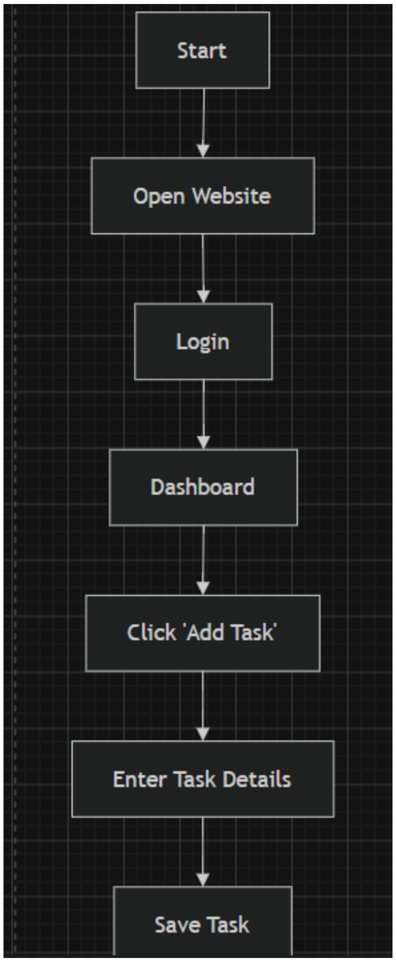
The task manager allows users to create, edit, delete, and prioritize tasks. It fulfills Functional Requirement 4.

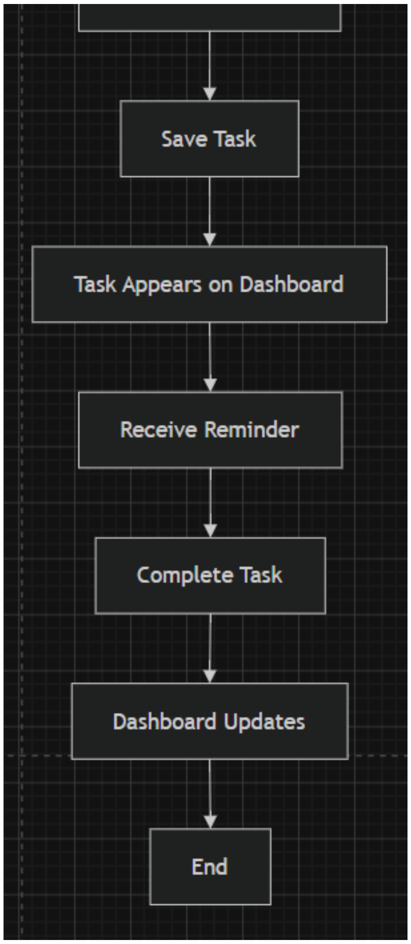
Wireframe 5 – Wellness Tracker

The wellness tracker helps users monitor healthy habits such as sleep, nutrition, exercise, and medications. It fulfills Functional Requirements 7 through 15.

3. User Flow Sequence

User Scenario: Creating and Completing a Task





User Flow Explanation

The user begins by opening the Arise application and logging into their account. After reaching the dashboard, they create a new task by entering the task details and saving it. The task is stored in the database and appears on the dashboard. When the due time approaches, the notification system reminds the user. Once the task is completed, the user marks it as complete, and the dashboard updates their progress.

Conclusion

The design of Arise focuses on creating a simple and organized platform for students to manage academics, work, and personal wellness. The system architecture, database design, wireframes, and user flow demonstrate how the application will function and provide an easy-to-use experience. This design serves as the foundation for the implementation phase of the project.